

Understanding Churn in Digital Subscriptions

News Website Case

Group 20.1

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A quick refresher...

We are analyzing churn to identify key drivers and improve subscriber retention for a news website.

Why does churn matter? Retention is more cost effective than acquisition. Understanding why users leave helps in allocating resources effectively.

Key Factors Analyzed:

- ◆ Trial offers
- ◆ Price
- ◆ Content engagement (regularity vs. intensity)
- ◆ Content Type
- ◆ Device Type

Data & Methodology

Dataset Overview



News website subscriber data
(One record per customer per month)

Methodology

- **Logistic Regression Models** — identifying factors influencing likelihood of churn
- **Model Validation** — Check for multicollinearity
- **Statistical significance** (p-values) to identify impactful variables

Dependent Var

Churn (1 = Canceled, 0 = Retained) → Whether a subscriber cancels their subscription next month

Independent Vars.

- **Regularity:** Days a user reads per month
- **Intensity:** Page views per reading day
- **Trial Status:** Whether the user is on a free trial
- **Next Price:** Amount paid next month
- **Content Consumption:** Sports, News, Crime, Business, Opinion, etc.
- **Device Usage:** Mobile, Tablet, Desktop sessions

Agenda

- **Regularity vs. Intensity:** Importance of Habit Formation
- Effect of **Time & Price** on Churn
- Why **Trial Offers** Are Ineffective
- How Regularity Carries Effects of **Content/Device Type**
- **DAG Analysis**
- **Recommended Strategies** by Customer Segment

Habit formation (regularity) matters more than binge-reading (intensity)

Let π = probability of churn

	Estimated Regression Equation	Analysis
Model 1: Regularity & Intensity	$\text{logit}(\pi) = -4.003(***) - 0.143t(***) + 0.36\text{trial}^* + 0.088\text{nextprice}^{***} - 0.027\text{regularity}(***) - 0.008\text{intensity}(\text{ns})$	Regularity remains significant (-0.027), while Intensity drops (-0.008) , suggests regularity is more crucial in decreasing likelihood of churn
Model 2: Regularity Only	$\text{logit}(\pi) = -4.049(***) - 0.14t(***) + 0.347\text{trial}^* + 0.087\text{nextprice}^{***} - 0.032\text{regularity}(***)$	Coefficient (-0.032) in regularity-only model is stronger than that (-0.019) in intensity-only model , indicates regularity reduces likelihood of churn more effectively than intensity
Model 3: Intensity Only	$\text{logit}(\pi) = -3.994(***) - 0.131t(***) + 0.325\text{trial}^* + 0.079\text{nextprice}^{***} - 0.019\text{intensity}(***)$	

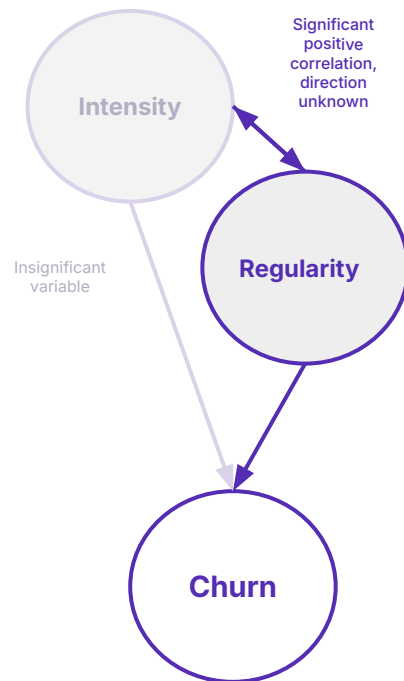
★ Takeaway

Reading more (intensity) doesn't improve retention **unless it is consistent (regularity)** → Focus on business strategies that encourage **consistent engagement & habit formation**

$\text{corr}(\text{regularity}, \text{intensity}) = 0.49^{***}$
→ **OVB** if any var removed

VIF regularity: 1.464
VIF intensity: 1.433
→ **no multicollinearity**

Directed Arc in a DAG



Controlling for other predictors, a longer customer life leads to lower likelihood of churn

Call:

```
glm(formula = nextchurn ~ t + trial + nextprice + regularity +  
     intensity, family = binomial, data = np)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-4.002903	0.353028	-11.339	< 2e-16 ***
t	-0.143106	0.029130	-4.913	8.98e-07 ***
trial	0.360129	0.155889	2.310	0.020879 *
nextprice	0.087507	0.018557	4.716	2.41e-06 ***
regularity	-0.026510	0.007067	-3.751	0.000176 ***
intensity	-0.007711	0.005163	-1.494	0.135285

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 3212.9 on 9534 degrees of freedom
Residual deviance: 3131.5 on 9529 degrees of freedom
(2635 observations deleted due to missingness)
AIC: 3143.5

Checking for VIF:

t
1.496

Context

Customers with a longer subscription history have already passed early churn risks

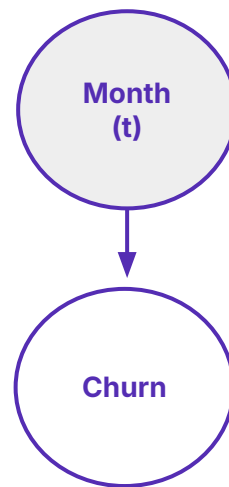
Why This Happens

Over time, customers build habits and integrate the service into their routines

★ Takeaway

Early-stage engagement strategies may help new customers reach this retention threshold

Directed Arc in a DAG



Controlling for other predictors, higher prices lead to higher likelihood of churn

```
Call:
glm(formula = nextchurn ~ t + trial + nextprice + regularity +
     intensity, family = binomial, data = np)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-4.002903	0.353028	-11.339	< 2e-16 ***
t	-0.143106	0.029130	-4.913	8.98e-07 ***
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AIC: 3143.5

Checking for VIF:

nextprice
1.035

Context

Higher prices may lead customers to reconsider their subscription

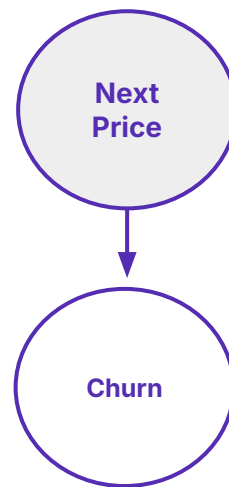
Why This Happens

Increased cost can reduce perceived value, prompting churn

★ Takeaway

Enhancing value perception of product may help improve retention

Directed Arc
in a DAG



Controlling for other predictors, customers who received a trial offer have a higher likelihood of churning

```
Call:
glm(formula = nextchurn ~ t + trial + nextprice + regularity +
    intensity, family = binomial, data = np)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-4.002903	0.353028	-11.339	< 2e-16 ***
t	-0.143106	0.029130	-4.913	8.98e-07 ***
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regularity	-0.026510	0.007067	-3.751	0.000176 ***
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Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 3212.9 on 9534 degrees of freedom
Residual deviance: 3131.5 on 9529 degrees of freedom
(2635 observations deleted due to missingness)
AIC: 3143.5

Checking for VIF:

trial
1.45

Context

One month trial period intended to drive conversion by lowering barrier to entry

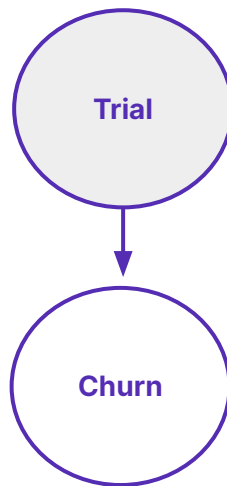
Why This Happens

First month of customer life very vulnerable & short trial period **limits habit formation**

★ Takeaway

Enhance **onboarding strategies for new customers** and **explore ways for regular engagement** (such as personalization)

Directed Arc
in a DAG



Regularity carries the effect of Content Type (News/ Sports) on Churn

Without Regularity:

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-4.117635	0.350838	-11.737	< 2e-16 ***
t	-0.130609	0.028793	-4.536	5.73e-06 ***
trial	0.309241	0.154966	1.996	0.0460 *
nextprice	0.083194	0.018471	4.504	6.67e-06 ***
sports1	-0.006065	0.002528	-2.399	0.0164 *
news1	-0.012748	0.005946	-2.144	0.0320 *
crime1	0.008753	0.007843	1.116	0.2644
life1	0.003819	0.008398	0.455	0.6493
obits1	-0.009301	0.013787	-0.675	0.4999
business1	-0.013241	0.026799	-0.494	0.6213
opinion1	0.026258	0.027764	0.946	0.3443

With Regularity:

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-4.0627006	0.3528121	-11.515	< 2e-16 ***
t	-0.1420869	0.0290650	-4.889	1.02e-06 ***
trial	0.3389870	0.1555876	2.179	0.02935 *
nextprice	0.0882696	0.0186122	4.743	2.11e-06 ***
sports1	-0.0006959	0.0027814	-0.250	0.80243
news1	-0.0087485	0.0057577	-1.519	0.12865
crime1	0.0108529	0.0076083	1.426	0.15373
life1	0.0046254	0.0079180	0.584	0.55911
obits1	-0.0012448	0.0137040	-0.091	0.92762
business1	-0.0094182	0.0265700	-0.354	0.72299
opinion1	0.0216623	0.0268220	0.808	0.41930
regularity	-0.0288405	0.0090123	-3.200	0.00137 **

	sports (regularity)		sports (w/o regularity)
VIF	1.76	⇒	1.15
Improves			
Without			
regularity	news (regularity)	⇒	news (w/o regularity)
	4.25		3.42

What We Tested

- Without regularity → News & Sports significantly reduce churn
- With regularity → News & Sports lose significance

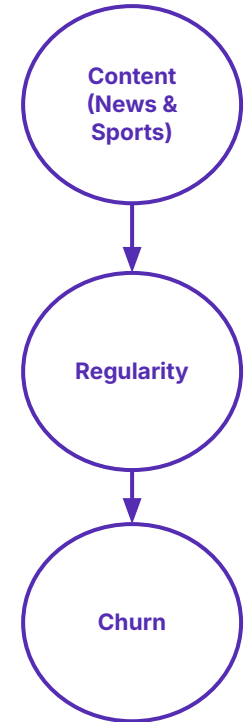
Why This Happens

- News & Sports content prone to daily viewing, (regularity), & regularity reduces churn
- Including both variables blocks indirect effect of content type on churn

★ What This Means

- Do not control for regularity when measuring content type's total effect
- Focus on promoting content that encourage habitual engagement

Pipe Structure in a DAG



Do not control for this as it blocks path

Regularity carries the effect of Device Type (Desktop) on Churn

Without Regularity:

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-4.086168	0.350129	-11.670	< 2e-16 ***
t	-0.129269	0.028701	-4.504	6.67e-06 ***
trial	0.307087	0.154737	1.985	0.0472 *
nextprice	0.083129	0.018413	4.515	6.34e-06 ***
mobile	-0.003066	0.002241	-1.368	0.1712
tablet	-0.007853	0.004065	-1.932	0.0534 .
desktop	-0.009112	0.002288	-3.983	6.80e-05 ***

With Regularity:

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-4.056118	0.351543	-11.538	< 2e-16 ***
t	-0.139194	0.028998	-4.800	1.59e-06 ***
trial	0.339821	0.155496	2.185	0.02886 *
nextprice	0.087613	0.018553	4.722	2.33e-06 ***
regularity	-0.028474	0.010969	-2.596	0.00943 **
mobile	0.001704	0.002761	0.617	0.53708
tablet	-0.001022	0.004529	-0.226	0.82147
desktop	-0.002576	0.003062	-0.841	0.40019

VIF
Improves
without
regularity

Desktop
(regularity)
2.31



Desktop
(w/o regularity)
1.04

What We Tested

- Without regularity → desktop viewing significantly reduces churn
- With regularity → desktop viewing loses significance

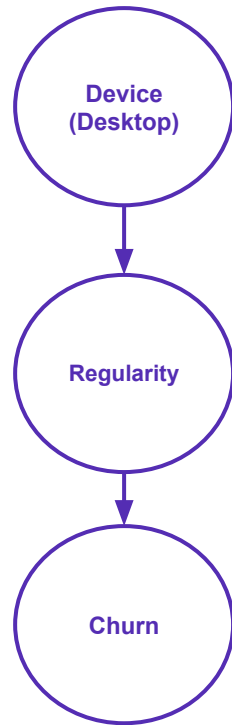
Why This Happens

- Desktop UI may be easier to read, leading **greater regularity**, & regularity decreases churn
- Including both variables blocks indirect effect of device type on churn**

★ What This Means

- Do not control for regularity** when measuring device type's total effect on churn likelihood
- Focus on maintaining desktop UI & improving other device experiences to encourage habitual engagement

Pipe Structure in a DAG



Do not control for this as it blocks path

Proving that Content & Device Type drive Regularity

Regressing Regularity on Content Type:

(Estimated) Regularity = 4.830 (***) + 0.131sports1 (***) + 0.103news1 (***) + 0.146crime1 (***) - 0.157life1 (***) + 0.328obits1 (***) + 0.125business1 (***) - 0.019opinion1 (***)

Given significance of content type predictors, we can conclude that **content type is a significant predictor of regularity**

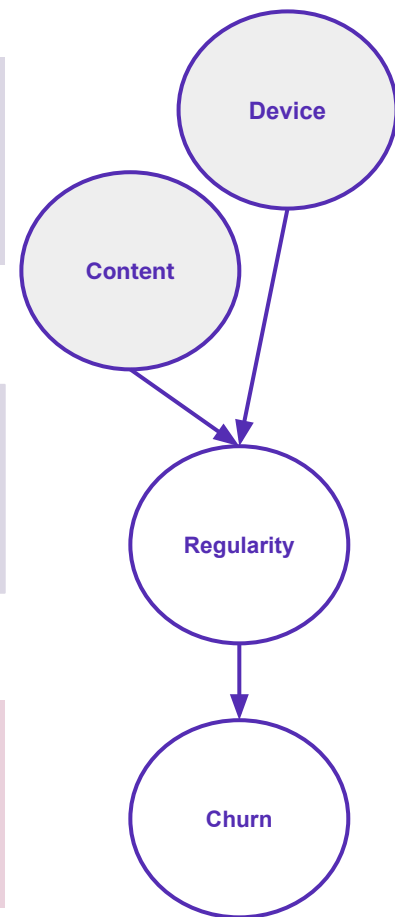
Regressing Regularity on Device Type:

(Estimated) Regularity = 3.466 (***) + 0.162mobile (***) + 0.222tablet (***) + 0.178desktop (***)

Given significance of device type predictors, we can conclude that **device type is a significant predictor of regularity**

★ Takeaways:

- **Regularity acts as a mediator** between content/device type on churn
- When controlling for regularity in the logit model, **the direct effect of content/device type on churn gets blocked or diminished**



Model with Payment, Content, Device Variables — Impact of Multicollinearity

full model:

$\text{logit}(\pi) = -4.0697 (***) - 0.1419t (***) + 0.3354\text{trial} (*) + 0.0886\text{nextprice} (***) - 0.0273\text{regularity} - \mathbf{0.0008\text{sports1}}$
 $- \mathbf{0.0088\text{news1}} + 0.0097\text{crime1} + 0.0053\text{life1} + 0.0006\text{obits1} - 0.0066\text{business1} + 0.0233\text{opinion1} + 0.0022\text{mobile} -$
 $0.0008\text{tablet} - \mathbf{0.0020\text{desktop} (*)}$

NextPrice & Device (Desktop) remain significant predictors that reduce likelihood of churn, but content type (Sports & News) no longer is

To check for multicollinearity, we conduct a correlation test between desktop & news/sports content

H0: $p=0$ there is no correlation between **desktop** & **sports** content

H1: $p \neq 0$ there is a significant correlation between desktop & sports content

$p\text{-value} < 2.2e-16 \rightarrow$ reject H0 & **conclude statistically significant correlation**

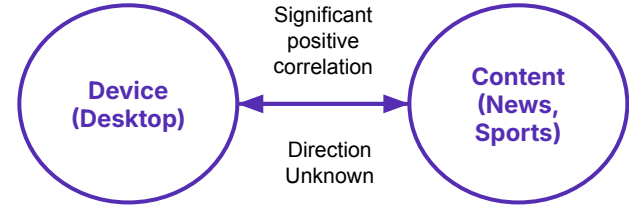
Corr(desktop, sports) = 0.51***

H0: $p=0$ there is no correlation between **desktop** & **news** content

H1: $p \neq 0$ there is a significant correlation between desktop & news content

$p\text{-value} < 2.2e-16 \rightarrow$ reject H0 & **conclude statistically significant correlation**

Corr(desktop, news) = 0.59***



- **Multicollinearity** between device type (desktop) and content type (news/sports) **affects regression results**
- **Significant correlation** means desktop users likely engage more with both content types, &/or news/sports may be better enjoyed on desktop
- **Overlap in predictive power** makes desktop significant, but content type insignificant in this model

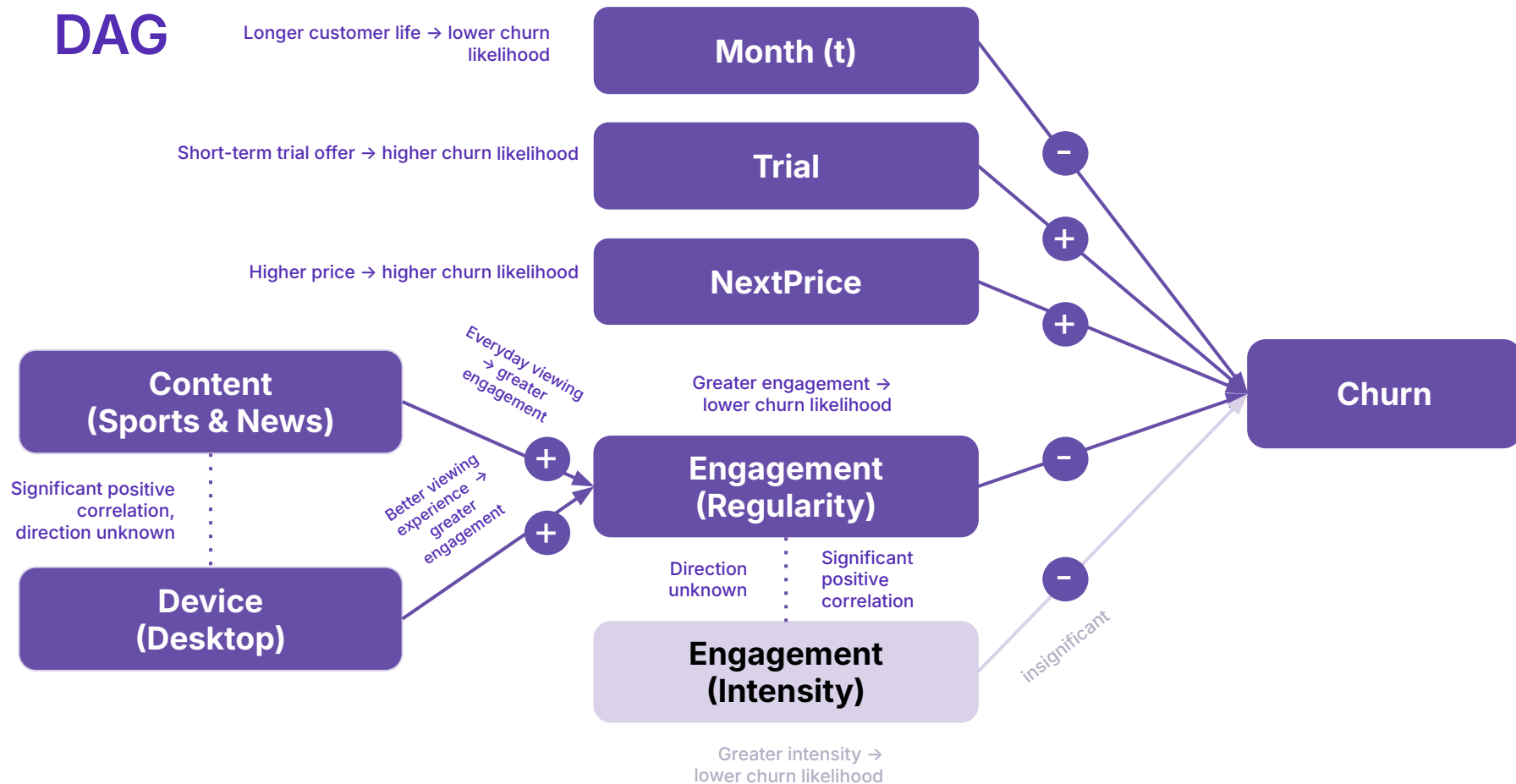
Checking for VIF:

Sports
1.730

News
4.031

Desktop
1.977

DAG



Recommended Strategies by Consumer Segment

 **Foster long-term customer loyalty & retention:** Focusing on strategies that encourage **regular engagement** rather than those that only incentivize short-term intensity (e.g. trial periods)

 **Create a sustainable customer lifecycle:** Initial Engagement → Habit Formation → Loyalty → Monetization

1. New & Casual Users (Acquisition & Initial Engagement)

 **Goal:** Attract new users and encourage regular engagement

 **Early Bird Offers** → Provide initial incentives (discounts, exclusive access) to encourage early and repeated interactions

 **Progressive Content Unlock with Personalized Algorithms** → Keep users engaged by offering content gradually, increasing their time spent and interest

Recommended Strategies by Consumer Segment

2. Returning & Habit-Building Users (Encouraging Consistency)

 **Goal:** Create habits and increase repeat visits

 **Gamification Elements (Challenges, Badges, Streaks, Leaderboards, Push Notifications)** → Motivate users to interact regularly by making engagement fun and rewarding while encourage social competition

3. Long-Term & Loyal Customers (Retention & Monetization)

 **Goal:** Strengthen loyalty and incentivize long-term commitment by **increasing value perception of product**

 **Loyalty Program (Points System & Rewards)** → Reward regular engagement and build deeper brand attachment

 **Subscription Model** → Convert loyal users into paying subscribers by offering exclusive perks and premium content

Questions or Feedback?

Let's discuss.